Q6): Library management system

Code:

```
<!DOCTYPE html>
<html>
<head>
<title>Library Management</title>
<style>
body{font-family:Arial;margin:0;background:#f1f5f9;color:#222}
header{background:#12345a;color:white;padding:18px;text-align:center}
nav{background:#087e8b;padding:12px;text-align:center}
nav a{color:white;margin:10px;text-decoration:none}
section{background:white;margin:20px;padding:20px;border-radius:10px}
input,button,textarea{padding:10px;margin:5px;border:1px solid #ccc;border-radius:5px}
button{background:#087e8b;color:white;cursor:pointer}
table{width:100%;border-collapse:collapse;margin-top:15px}
th,td{padding:10px;border:1px solid #ddd;text-align:left}
th{background:#12345a;color:white}
</style>
</head>
<body>

<header>
<h1>📚 Library Management System</h1>
<p>Search • Issue • Return • Manage Books</p>
</header>

<nav>
<a href="#register">Register</a>
<a href="#login">Login</a>
<a href="#books">Books</a>
<a href="#admin">Admin</a>
<a href="#fine">Fine</a>
```

```
<a href="#contact">Contact</a>
</nav>

<section id="register">
<h2>Student Registration</h2>
<input id="regName" placeholder="Full Name">
<input id="regEmail" type="email" placeholder="Email">
<input id="regPass" type="password" placeholder="Password">
<button onclick="register()">Register</button>
<p id="regMsg"></p>
</section>

<section id="login">
<h2>Student Login</h2>
<input id="student" type="email" placeholder="Registered Email">
<input id="loginPass" type="password" placeholder="Password">
<button onclick="login()">Login</button>
<p id="welcome"></p>
</section>

<section id="books">
<h2>Book Catalog</h2>
<input id="search" placeholder="Search book or author" oninput="showBooks()">
<table>
<thead><tr><th>Book</th><th>Author</th><th>Status</th><th>Action</th></tr></thead>
<tbody id="bookList"></tbody>
</table>
</section>

<section id="admin">
<h2>Admin Dashboard</h2>
<p id="count"></p>
```

```
<button onclick="addBook()">Add Book</button>
</section>

<section id="fine">
<h2>Fine Calculator</h2>
<input type="number" id="days" placeholder="Overdue days" min="0">
<button onclick="fine()">Calculate Fine</button>
<p id="result"></p>
</section>

<section id="contact">
<h2>Contact Us</h2>
<input id="name" placeholder="Your name">
<input id="email" type="email" placeholder="Your email"><br>
<textarea id="message" placeholder="Your message"></textarea><br>
<button onclick="contact()">Submit</button>
<p id="reply"></p>
</section>

<script>
let user="";
let users=[];
let books=[
{title:"Wings of Fire",author:"A.P.J. Abdul Kalam",available:true},
{title:"The Alchemist",author:"Paulo Coelho",available:true},
{title:"India After Gandhi",author:"Ramachandra Guha",available:false}
];

function register(){
 let name=document.getElementById("regName").value.trim();
 let email=document.getElementById("regEmail").value.trim().toLowerCase();
 let pass=document.getElementById("regPass").value;
```

```
 if(!name||!email.includes("@")||pass.length<4){
  alert("Enter valid details. Password needs 4 characters.");
  return;
 }
 if(users.some(u=>u.email===email)){
  alert("Email already registered!");
  return;
 }
 users.push({name,email,pass});
 document.getElementById("regMsg").innerText=
 "Registration successful! Please login.";
}

function login(){
 let email=document.getElementById("student").value.trim().toLowerCase();
 let pass=document.getElementById("loginPass").value;
 let found=users.find(u=>u.email===email&&u.pass===pass);

 if(found){
  user=found.name;
  document.getElementById("welcome").innerText="Welcome, "+user;
 }else{
  alert("Invalid details. Please register first.");
 }
}

function showBooks(){
 let q=document.getElementById("search").value.toLowerCase();
 let list=document.getElementById("bookList");
 list.innerHTML="";
 books.forEach((b,i)=>{
```

```
  if((b.title+" "+b.author).toLowerCase().includes(q)){
   list.innerHTML+=`<tr><td>${b.title}</td><td>${b.author}</td>
   <td>${b.available?"Available":"Issued"}</td>
   <td><button onclick="changeBook(${i})">${b.available?"Issue":"Return"}</button></td></tr>`;
  }
 });
 document.getElementById("count").innerText=
 "Total Books: "+books.length+" | Available: "+
 books.filter(b=>b.available).length;
}

function changeBook(i){
 if(!user){alert("Please login first");return;}
 books[i].available=!books[i].available;
 showBooks();
 alert("Book status updated!");
}

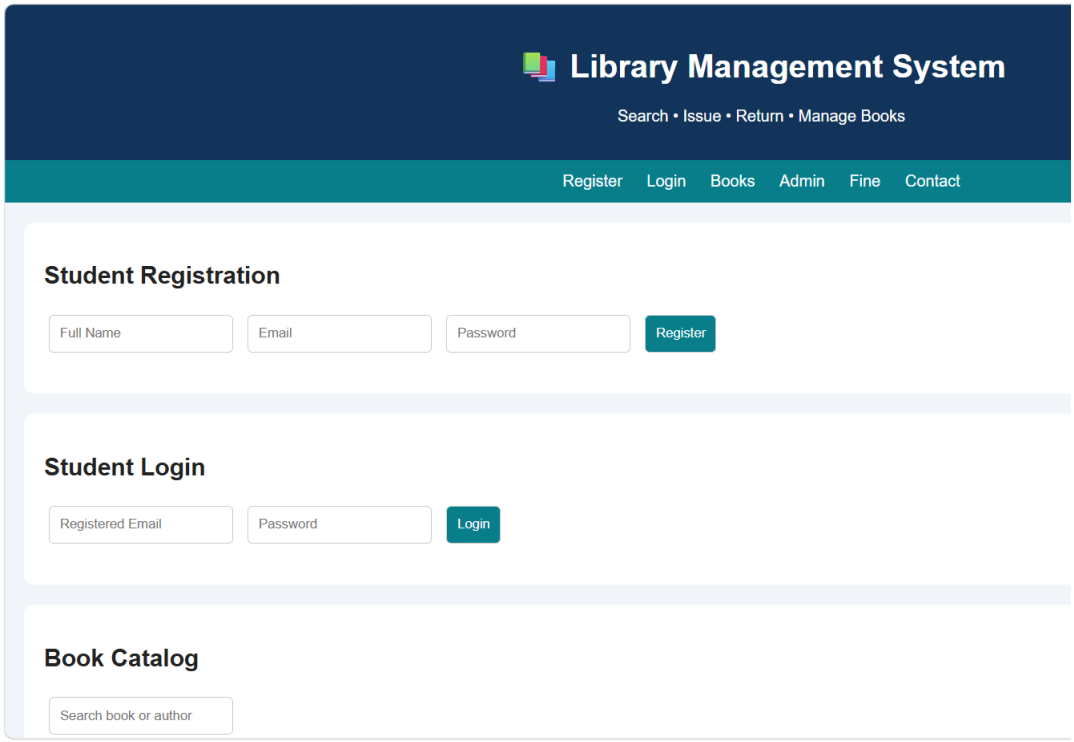
function addBook(){
 let title=prompt("Enter book name:");
 if(!title||!title.trim())return;
 let author=prompt("Enter author name:")||"Unknown";
 books.push({title:title.trim(),author:author,available:true});
 showBooks();
}

function fine(){
 let d=Math.max(0,Number(document.getElementById("days").value)||0);
 document.getElementById("result").innerText=
 "Total Fine: ₹"+(d*5)+" (₹5 per overdue day)";
}
```

```
function contact(){
 let n=document.getElementById("name").value.trim();
 let e=document.getElementById("email").value.trim();
 let m=document.getElementById("message").value.trim();
 document.getElementById("reply").innerText=
 n&&e.includes("@")&&m?"Thank you! Message submitted.":"Please enter valid details.";
}

showBooks();
</script>
</body>
</html>
```

Output:



Q2) Hotel room booking

Code:

```
<!DOCTYPE html>
<html>
<head>
<title>Hotel Booking System</title>
```

```
<style>
body{font-family:Arial;margin:0;background:#f1f5f9;color:#222}
header{background:#12345a;color:white;padding:20px;text-align:center}
nav{background:#087e8b;padding:12px;text-align:center}
nav a{color:white;margin:8px;text-decoration:none}
section{background:white;margin:18px;padding:18px;border-radius:10px}
input,select,button,textarea{padding:10px;margin:5px;border:1px solid #ccc;border-radius:5px}
button{background:#087e8b;color:white;cursor:pointer}
.room{display:inline-block;background:#e8f4f5;padding:15px;margin:8px;border-radius:8px}
</style>
</head>
<body>

<header>
<h1>🏨 Grand Stay Hotel</h1>
<p>Comfort • Luxury • Affordable Prices</p>
</header>

<nav>
<a href="#register">Register</a><a href="#login">Login</a>
<a href="#rooms">Rooms</a><a href="#booking">Book</a>
<a href="#history">History</a><a href="#admin">Admin</a>
<a href="#contact">Contact</a>
</nav>

<section id="register">
<h2>User Registration</h2>
<input id="rname" placeholder="Full Name">
<input id="remail" type="email" placeholder="Email">
<input id="rpass" type="password" placeholder="Password">
<button onclick="register()">Register</button>
<p id="rmsg"></p>
```

```
</section>

<section id="login">
<h2>Login</h2>
<input id="lemail" type="email" placeholder="Registered Email">
<input id="lpass" type="password" placeholder="Password">
<button onclick="login()">Login</button>
<p id="welcome"></p>
</section>

<section id="rooms">
<h2>Available Room Types</h2>
<div id="roomList"></div>
</section>

<section id="booking">
<h2>Search & Book a Room</h2>
<input id="destination" placeholder="Hotel destination">
<select id="type">
<option>Standard</option><option>Deluxe</option>
<option>Suite</option><option>Family</option>
</select><br>
<label>Check-in:</label><input type="date" id="checkin">
<label>Check-out:</label><input type="date" id="checkout">
<input type="number" id="guests" min="1" value="1" placeholder="Guests">
<button onclick="bookRoom()">Check & Book</button>
<p id="bookingMsg"></p>
</section>

<section id="history">
<h2>My Booking History</h2>
<div id="myBookings">Login to view bookings.</div>
```

```
</section>

<section id="admin">
<h2>Admin Panel</h2>
<p>Demo admin access: username <b>admin</b>, password <b>admin123</b></p>
<input id="aname" placeholder="Admin username">
<input id="apass" type="password" placeholder="Admin password">
<button onclick="adminLogin()">Admin Login</button>
<div id="adminPanel" style="display:none">
<h3>Manage Room Price & Quantity</h3>
<select id="aroom">
<option>Standard</option><option>Deluxe</option>
<option>Suite</option><option>Family</option>
</select><br>
<input id="price" type="number" placeholder="Price per night">
<input id="quantity" type="number" placeholder="Room quantity">
<button onclick="updateRoom()">Update Room</button>
<h3>Manage Bookings</h3>
<div id="allBookings"></div>
</div>
</section>

<section id="info">
<h2>About Us</h2>
<p>Grand Stay provides comfortable rooms at affordable prices.</p>
<h3>Customer Service FAQs</h3>
<p>Q: Can I cancel? A: Yes, up to 3 days before check-in.</p>
<p>Q: Is payment available? A: This demo does not process real payments.</p>
<h3>Local Travel Guide</h3>
<p>Ask our reception for nearby attractions and transport information.</p>
<h3>Privacy Policy</h3>
<p>Do not enter sensitive personal or payment details in this demo.</p>
```

```
</section>

<section id="contact">
<h2>Contact Us</h2>
<p>Email: support@grandstay.example | Phone: 9876543210</p>
<input id="cname" placeholder="Your name">
<input id="cemail" type="email" placeholder="Your email"><br>
<textarea id="cmsg" placeholder="Your message"></textarea><br>
<button onclick="contact()">Submit</button>
<p id="cresult"></p>
</section>

<script>
let users=[],current=null,bookings=[];
let rooms={
 Standard:{price:1500,qty:5},
 Deluxe:{price:2500,qty:4},
 Suite:{price:4000,qty:2},
 Family:{price:3500,qty:3}
};

function register(){
 let name=rname.value.trim(),email=remail.value.trim().toLowerCase(),pass=rpass.value;
 if(!name||!email.includes("@")||pass.length<4)
 return alert("Enter valid details and password (4+ characters).");
 if(users.some(u=>u.email===email))return alert("Email already registered.");
 users.push({name,email,pass});
 rmsg.innerText="Registration successful! Please login.";
}

function login(){
 let email=lemail.value.trim().toLowerCase();
```

```
 let u=users.find(x=>x.email===email&&x.pass===lpass.value);
 if(!u)return alert("Invalid login. Please register first.");
 current=u;
 welcome.innerText="Welcome, "+u.name;
 showBookings();
}

function showRooms(){
 roomList.innerHTML="";
 for(let t in rooms){
  roomList.innerHTML+=`<div class="room"><h3>${t}</h3>
  <p>₹${rooms[t].price} per night</p>
  <p>Rooms: ${rooms[t].qty}</p></div>`;
 }
}
function bookRoom(){
 if(!current)return alert("Please login first.");
 let start=checkin.value,end=checkout.value,t=type.value;
 if(!destination.value.trim()||!start||!end||!guests.value)
 return alert("Fill in all booking details.");
 let a=new Date(start+"T00:00:00"),b=new Date(end+"T00:00:00");
 let today=new Date();today.setHours(0,0,0,0);
 if(a<today||b<=a)return alert("Select valid future dates.");
 let nights=(b-a)/86400000;
 let used=bookings.filter(x=>x.type===t&&x.status==="Booked"&&
 x.start<end&&x.end>start).length;
 if(used>=rooms[t].qty)return alert("No rooms available for these dates.");
 let total=nights*rooms[t].price;
 bookings.push({id:Date.now(),email:current.email,type:t,
 start,end,guests:guests.value,total,status:"Booked"});
 bookingMsg.innerText="Booking confirmed! Total: ₹"+total;
 showBookings();showAdminBookings();
```

```
}

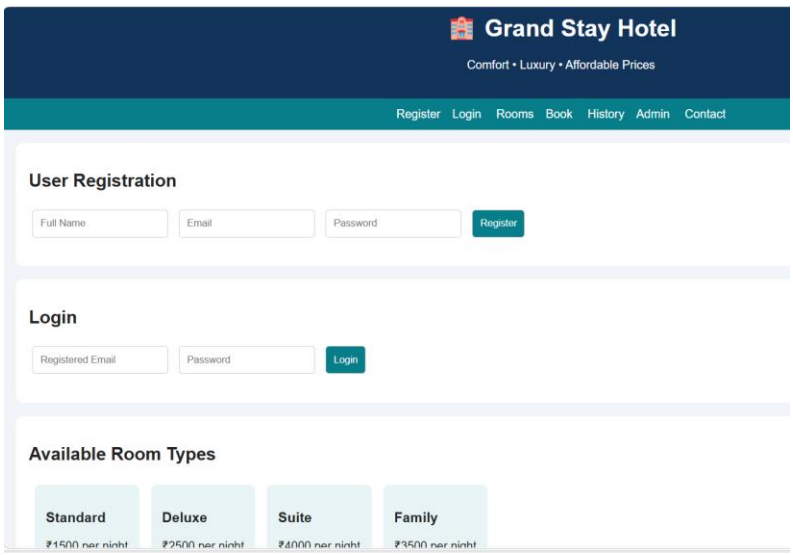
function showBookings(){
 let mine=bookings.filter(b=>b.email===current.email);
 myBookings.innerHTML=mine.length?"":"No bookings yet.";
 mine.forEach(b=>{
 myBookings.innerHTML+=`<p>${b.type} | ${b.start} to ${b.end} |
 ₹${b.total} | ${b.status}
 ${b.status==="Booked"?`<button onclick="cancelBooking(${b.id})">Cancel</button>
 <button onclick="changeDate(${b.id})">Change Dates</button>`:""}</p>`;
 });
}
function cancelBooking(id){
 let b=bookings.find(x=>x.id===id);
 let limit=new Date(b.start+"T00:00:00");
 limit.setDate(limit.getDate()-3);
 if(new Date()>=limit)return alert("Cancellation allowed only 3+ days before arrival.");
 b.status="Cancelled";showBookings();showAdminBookings();showRooms();
}
function changeDate(id){
 let b=bookings.find(x=>x.id===id);
 let limit=new Date(b.start+"T00:00:00");
 limit.setDate(limit.getDate()-3);
 if(new Date()>=limit)return alert("Changes allowed only 3+ days before arrival.");
 let s=prompt("New check-in date (YYYY-MM-DD)",b.start);
 let e=prompt("New check-out date (YYYY-MM-DD)",b.end);
 if(!s||!e||e<=s||s<new Date().toLocaleDateString("en-CA"))
 return alert("Invalid dates.");
 let
used=bookings.filter(x=>x.id!==id&&x.type===b.type&&x.status==="Booked"&&x.start<e&&x.end>s).length;
 if(used>=rooms[b.type].qty)return alert("No rooms available on these dates.");
 b.start=s;b.end=e;
```

```
 b.total=((new Date(e)-new Date(s))/86400000)*rooms[b.type].price;
 showBookings();showAdminBookings();
}

function adminLogin(){
 if(aname.value==="admin"&&apass.value==="admin123"){
 adminPanel.style.display="block";showAdminBookings();
 }else alert("Invalid admin login.");
}
function updateRoom(){
 let t=aroom.value,p=Number(price.value),q=Number(quantity.value);
 if(p<=0||q<0||!price.value||!quantity.value)
 return alert("Enter valid price and quantity.");
 rooms[t].price=p;rooms[t].qty=q;
 showRooms();showAdminBookings();
 alert("Room details updated!");
}
function showAdminBookings(){
 allBookings.innerHTML="";
 bookings.forEach(b=>{
 allBookings.innerHTML+=`<p>#${b.id} | ${b.email} | ${b.type} |
 ${b.start} - ${b.end} | ${b.status}
 <button onclick="adminCancel(${b.id})">Cancel</button>
 <button onclick="adminChange(${b.id})">Edit Dates</button></p>`;
 });
}
function adminCancel(id){
 let b=bookings.find(x=>x.id===id);
 b.status="Cancelled";showBookingsIfCurrent();showAdminBookings();showRooms();
}
function adminChange(id){
 let b=bookings.find(x=>x.id===id);
```

```
 let s=prompt("New check-in date (YYYY-MM-DD)",b.start);
 let e=prompt("New check-out date (YYYY-MM-DD)",b.end);
 if(!s||!e||e<=s)return alert("Invalid dates.");
 b.start=s;b.end=e;
 b.total=((new Date(e)-new Date(s))/86400000)*rooms[b.type].price;
 showBookingsIfCurrent();showAdminBookings();
}
function showBookingsIfCurrent(){if(current)showBookings();}
function contact(){
 cresult.innerText=cname.value.trim()&&cemail.value.includes("@")&&cmsg.value.trim()
 ?"Thank you! Message submitted.":"Enter valid details.";
}
showRooms();
</script>
</body>
</html>
```

Output:



Q3) Online bus ticket booking system

Code:

```
<!DOCTYPE html>
<html>
<head>
<title>Bus Ticket Booking</title>
<style>
*{box-sizing:border-box}
body{font-family:Arial;margin:0;background:#f4f6fb;color:#263248}
header{background:#172554;color:white;padding:25px;text-align:center}
header h1{margin:5px}
nav{background:#f97316;padding:14px;text-align:center}
nav a{color:white;text-decoration:none;margin:10px;font-weight:bold}
section{background:white;margin:20px auto;padding:22px;border-radius:12px;
max-width:1000px;box-shadow:0 3px 12px #ddd}
h2{color:#172554}
input,select,textarea,button{padding:11px;margin:5px;border:1px solid #ccc;border-radius:6px}
button{background:#f97316;color:white;border:0;cursor:pointer}
button:hover{background:#ea580c}
.bus{background:#eef2ff;padding:15px;margin:10px 0;border-radius:8px}
.seats{display:flex;flex-wrap:wrap;gap:8px;margin:12px}
.seat{background:#16a34a;padding:12px;color:white;border-radius:5px;cursor:pointer}
.seat.selected{background:#f97316}
.seat.booked{background:#777;cursor:not-allowed}
table{width:100%;border-collapse:collapse}
td,th{padding:10px;border:1px solid #ddd;text-align:left}
th{background:#172554;color:white}
footer{text-align:center;background:#172554;color:white;padding:18px}
</style>
</head>
<body>

<header>
<h1>🚌 QuickBus</h1>
```

```
<p>Book Your Journey Easily and Safely</p>
</header>

<nav>
<a href="#register">Register</a>
<a href="#login">Login</a>
<a href="#search">Search Buses</a>
<a href="#history">My Tickets</a>
<a href="#admin">Admin</a>
<a href="#comments">Comments</a>
</nav>

<section id="register">
<h2>Create Account</h2>
<input id="rname" placeholder="Full Name">
<input id="remail" type="email" placeholder="Email">
<input id="rpass" type="password" placeholder="Password">
<button onclick="register()">Register</button>
<p id="rmsg"></p>
</section>

<section id="login">
<h2>User Login</h2>
<input id="lemail" type="email" placeholder="Registered Email">
<input id="lpass" type="password" placeholder="Password">
<button onclick="login()">Login</button>
<button onclick="forgot()">Forgot Password</button>
<p id="welcome"></p>
</section>

<section id="search">
<h2>Find Your Bus</h2>
```

```
<input id="from" placeholder="From city">
<input id="to" placeholder="To city">
<input id="date" type="date">
<button onclick="searchBuses()">Search Buses</button>
<div id="results"></div>
</section>

<section id="booking">
<h2>Passenger & Seat Booking</h2>
<p id="selectedBus">Select a bus from search results.</p>
<div id="seatArea" class="seats"></div>
<input id="passenger" placeholder="Passenger Name">
<input id="age" type="number" placeholder="Age">
<button onclick="bookTicket()">Confirm Booking</button>
<p id="bookingMsg"></p>
</section>

<section id="history">
<h2>My Tickets</h2>
<div id="tickets">Login to view your tickets.</div>
</section>

<section id="admin">
<h2>Admin Panel</h2>
<p>Demo login: admin / admin123</p>
<input id="aname" placeholder="Admin username">
<input id="apass" type="password" placeholder="Admin password">
<button onclick="adminLogin()">Admin Login</button>
<div id="adminPanel" style="display:none">
<h3>Add Bus</h3>
<input id="busname" placeholder="Bus name">
<input id="category" placeholder="Category (Volvo/Sleeper)">
```

```
<input id="routefrom" placeholder="From">
<input id="routeto" placeholder="To">
<input id="time" type="time">
<input id="fare" type="number" placeholder="Ticket price">
<input id="capacity" type="number" placeholder="Number of seats">
<button onclick="addBus()">Add Bus</button>
<h3>Bus Information & Reports</h3>
<div id="report"></div>
<button onclick="generateReport()">Generate Report</button>
</div>
</section>

<section id="comments">
<h2>Passenger Comments</h2>
<input id="commentName" placeholder="Your name">
<textarea id="commentText" placeholder="Write a comment"></textarea>
<button onclick="addComment()">Post Comment</button>
<div id="commentList"></div>
</section>

<footer>
<p>QuickBus Ticket Booking System | Internet Programming Lab</p>
<p>Contact: support@quickbus.example</p>
</footer>

<script>
let users=[],current=null,selected=null,chosen=[];
let buses=[
{id:1,name:"Express Travels",category:"Volvo",from:"Hyderabad",to:"Vijayawada",time:"09:00",fare:650,capacity:8,booked:[]},
{id:2,name:"Night Rider",category:"Sleeper",from:"Hyderabad",to:"Warangal",time:"21:00",fare:450,capacity:8,booked:[]},
```

```
{id:3,name:"City
Travels",category:"Coach",from:"Vijayawada",to:"Hyderabad",time:"14:00",fare:550,capacity:8,book
ed:[]}
];
let tickets=[],comments=[];

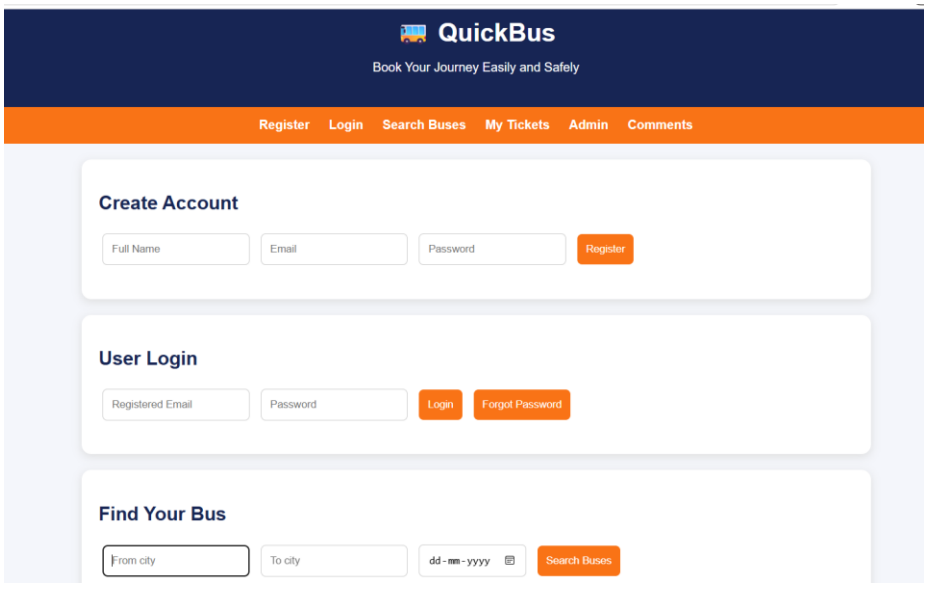
function register(){
 let name=rname.value.trim(),email=remail.value.trim().toLowerCase(),pass=rpass.value;
 if(!name||!email.includes("@")||pass.length<4)return alert("Enter valid details.");
 if(users.some(u=>u.email===email))return alert("Email already registered.");
 users.push({name,email,pass});
 rmsg.innerText="Registration successful! Please login.";
}
function login(){
 let email=lemail.value.trim().toLowerCase();
 current=users.find(u=>u.email===email&&u.pass===lpass.value);
 if(current){welcome.innerText="Welcome, "+current.name;showTickets();}
 else{current=null;alert("Invalid login. Register first.");}
}
function forgot(){
 let email=prompt("Enter your registered email:");
 let u=users.find(x=>x.email===email?.trim().toLowerCase());
 alert(u?"Demo: Contact admin to reset your password.":"Email not found.");
}
function searchBuses(){
 results.innerHTML="";
 let f=from.value.trim().toLowerCase(),t=to.value.trim().toLowerCase();
 buses.filter(b=>b.from.toLowerCase().includes(f)&&b.to.toLowerCase().includes(t))
 .forEach(b=>{
 results.innerHTML+=`<div class="bus"><h3>${b.name} - ${b.category}</h3>
 <p>${b.from} → ${b.to} | ${b.time} | ₹${b.fare}</p>
 <p>Available seats: ${b.capacity-b.booked.filter(x=>x.date===date.value).length}</p>
 <button onclick="selectBus(${b.id})">Select Bus</button></div>`;
```

```
 });
 if(!results.innerHTML)results.innerText="No buses found for this route.";
}
function selectBus(id){
 if(!date.value)return alert("Select journey date first.");
 if(date.value<new Date().toLocaleDateString("en-CA"))return alert("Select a valid date.");
 selected=buses.find(b=>b.id===id);chosen=[];
 selectedBus.innerText=selected.name+" | ₹"+selected.fare+" per seat";
 showSeats();
}
function showSeats(){
 seatArea.innerHTML="";
 for(let i=1;i<=selected.capacity;i++){
 let occupied=selected.booked.some(x=>x.seat===i&&x.date===date.value);
 seatArea.innerHTML+=`<button class="seat ${occupied?"booked":""}
${chosen.includes(i)?"selected":""}"
 ${occupied?'disabled':`onclick="toggleSeat(${i})"`}>${i}</button>`;
 }
}
function toggleSeat(n){
 chosen.includes(n)?chosen=chosen.filter(x=>x!==n):chosen.push(n);
 showSeats();
}
function bookTicket(){
 if(!current)return alert("Please login first.");
 if(!selected||!chosen.length)return alert("Select a bus and at least one seat.");
 if(!passenger.value.trim()||Number(age.value)<1)return alert("Enter passenger details.");
 if(chosen.some(n=>selected.booked.some(x=>x.seat===n&&x.date===date.value)))
 return alert("A selected seat is no longer available.");
 let ticket={id:Date.now(),email:current.email,name:passenger.value.trim(),
 bus:selected.name,from:selected.from,to:selected.to,date:date.value,
 seats:[...chosen],total:selected.fare*chosen.length,status:"Confirmed"};
 chosen.forEach(n=>selected.booked.push({seat:n,date:date.value}));
```

```
 tickets.push(ticket);
 bookingMsg.innerText="Booking confirmed! Total: ₹"+ticket.total;
 chosen=[];showSeats();showTickets();searchBuses();
}
function showTickets(){
 let own=tickets.filter(t=>t.email===current.email);
 ticketsDiv(own);
}
function ticketsDiv(list){
 tickets.innerHTML=list.length?"":"No tickets found.";
 list.forEach(t=>{
 tickets.innerHTML+=`<div class="bus">${t.bus}: ${t.from} → ${t.to}<br>
 Date: ${t.date} | Seats: ${t.seats.join(", ")} | ₹${t.total}
 | ${t.status} ${t.status==="Confirmed"?`<button
onclick="cancelTicket(${t.id})">Cancel</button>`:""}
 </div>`;
 });
}
function cancelTicket(id){
 let t=tickets.find(x=>x.id===id);
 if(t.status!=="Confirmed")return;
 let d=new Date(t.date+"T00:00:00");let today=new Date();today.setHours(0,0,0,0);
 if(d<today)return alert("Past journey tickets cannot be cancelled in this demo.");
 let b=buses.find(x=>x.name===t.bus);
 b.booked=b.booked.filter(x=>!(t.seats.includes(x.seat)&&x.date===t.date));
 t.status="Cancelled";showTickets();searchBuses();
}
function adminLogin(){
 if(aname.value==="admin"&&apass.value==="admin123"){
 adminPanel.style.display="block";generateReport();
 }else alert("Invalid admin login.");
}
function addBus(){
```

```
 let name=busname.value.trim(),f=routefrom.value.trim(),t=routeto.value.trim();
 let cap=Number(capacity.value),price=Number(fare.value);
 if(!name||!f||!t||cap<1||price<1)return alert("Enter valid bus details.");
 buses.push({id:Date.now(),name,category:category.value||"Coach",
 from:f,to:t,time:time.value||"09:00",fare:price,capacity:cap,booked:[]});
 alert("Bus added successfully!");generateReport();
}
function generateReport(){
 report.innerHTML=`Total buses: ${buses.length}<br>Total tickets: ${tickets.length}<br>
 Confirmed: ${tickets.filter(t=>t.status==="Confirmed").length}<br>
 Cancelled: ${tickets.filter(t=>t.status==="Cancelled").length}<br>
 Revenue: ₹${tickets.filter(t=>t.status==="Confirmed").reduce((s,t)=>s+t.total,0)}`;
}
function addComment(){
 if(!commentName.value.trim()||!commentText.value.trim())return alert("Enter name and comment.");
 comments.push({name:commentName.value.trim(),text:commentText.value.trim()});
 commentList.innerHTML=comments.map(c=>`<p><b>${c.name}</b>: ${c.text}</p>`).join("");
}
</script>
</body>
</html>
```

Output:



Q4): **Online train reservation system**

Code:

```
<!DOCTYPE html>
<html>
<head>
<title>Railway Reservation System</title>
<style>
*{box-sizing:border-box}
body{font-family:Arial;margin:0;background:#f5f3ff;color:#292744}
header{background:linear-gradient(120deg,#312e81,#6366f1);
color:white;text-align:center;padding:25px}
nav{background:#1e1b4b;padding:14px;text-align:center}
nav a{color:white;text-decoration:none;margin:9px;font-weight:bold}
section{background:white;margin:20px auto;padding:22px;
max-width:1000px;border-radius:14px;box-shadow:0 4px 15px #ddd}
h2{color:#4338ca}
input,select,button{padding:10px;margin:5px;
border:1px solid #ccc;border-radius:6px}
button{background:#4f46e5;color:white;border:0;cursor:pointer}
button:hover{background:#3730a3}
.train{background:#eef2ff;padding:15px;margin:12px 0;border-radius:10px}
table{width:100%;border-collapse:collapse}
td,th{padding:10px;border:1px solid #ddd;text-align:left}
th{background:#312e81;color:white}
footer{background:#1e1b4b;color:white;text-align:center;padding:18px}
</style>
</head>
<body>

<header>
<h1>🚆 RailConnect</h1>
```

```
<p>Railway Reservation and Ticket Management System</p>
</header>

<nav>
<a href="#register">Register</a>
<a href="#login">Login</a>
<a href="#search">Find Trains</a>
<a href="#booking">Book Ticket</a>
<a href="#pnr">PNR Status</a>
<a href="#admin">Admin</a>
</nav>

<section id="register">
<h2>Passenger Registration</h2>
<input id="rname" placeholder="Full Name">
<input id="remail" type="email" placeholder="Email">
<input id="rpass" type="password" placeholder="Password">
<button onclick="register()">Register</button>
<p id="rmsg"></p>
</section>

<section id="login">
<h2>Passenger Login</h2>
<input id="lemail" type="email" placeholder="Registered Email">
<input id="lpass" type="password" placeholder="Password">
<button onclick="login()">Login</button>
<p id="welcome"></p>
</section>

<section id="search">
<h2>Search Trains</h2>
<input id="origin" placeholder="From Station">
```

```
<input id="destination" placeholder="To Station">
<input id="journeyDate" type="date">
<select id="classType">
<option>Sleeper</option>
<option>AC 3 Tier</option>
<option>AC 2 Tier</option>
<option>First Class</option>
</select>
<button onclick="searchTrains()">Search Trains</button>
<div id="trainList"></div>
</section>

<section id="booking">
<h2>Passenger & Ticket Booking</h2>
<p id="selectedTrain">Select a train from the search results.</p>
<input id="passengerName" placeholder="Passenger Name">
<input id="passengerAge" type="number" placeholder="Age">
<select id="payment">
<option>Online (Demo)</option>
<option>Offline (Demo)</option>
</select>
<button onclick="bookTicket()">Confirm Booking</button>
<p id="bookingMsg"></p>
<div id="ticket"></div>
</section>

<section id="pnr">
<h2>Check PNR Status</h2>
<input id="pnrInput" placeholder="Enter PNR Number">
<button onclick="checkPNR()">Check Status</button>
<p id="pnrResult"></p>
</section>
```

```
<section id="history">
<h2>My Reservations</h2>
<div id="historyList">Login to view your reservations.</div>
</section>

<section id="cancel">
<h2>Cancel Ticket & Refund</h2>
<input id="cancelPNR" placeholder="Enter PNR Number">
<button onclick="cancelTicket()">Cancel Ticket</button>
<p id="refundResult"></p>
</section>

<section id="admin">
<h2>Administrator Panel</h2>
<p>Demo login: admin / admin123</p>
<input id="adminName" placeholder="Admin Username">
<input id="adminPass" type="password" placeholder="Admin Password">
<button onclick="adminLogin()">Admin Login</button>

<div id="adminPanel" style="display:none">
<h3>Add Train</h3>
<input id="newTrain" placeholder="Train Name">
<input id="newFrom" placeholder="Origin">
<input id="newTo" placeholder="Destination">
<input id="newFare" type="number" placeholder="Fare">
<input id="newSeats" type="number" placeholder="Available Seats">
<button onclick="addTrain()">Add Train</button>

<h3>Reservation Report</h3>
<div id="report"></div>
<button onclick="generateReport()">Refresh Report</button>
```

```
</div>
</section>

<footer>
<p>RailConnect | Internet Programming Lab Experiment</p>
<p>Train reservations made simple.</p>
</footer>

<script>
let users=[],current=null,selected=null,tickets=[];
let trains=[
{id:1,name:"Express Superfast",from:"Hyderabad",to:"Chennai",fare:650,seats:5},
{id:2,name:"Deccan Express",from:"Hyderabad",to:"Vijayawada",fare:400,seats:8},
{id:3,name:"Coastal Express",from:"Chennai",to:"Vijayawada",fare:500,seats:6}
];

function register(){
 let name=rname.value.trim(),email=remail.value.trim().toLowerCase(),pass=rpass.value;
 if(!name||!email.includes("@")||pass.length<4)
 return alert("Enter valid details. Password needs 4 characters.");
 if(users.some(u=>u.email===email))return alert("Email already registered.");
 users.push({name,email,pass});
 rmsg.innerText="Registration successful! Please login.";
}

function login(){
 let email=lemail.value.trim().toLowerCase();
 current=users.find(u=>u.email===email&&u.pass===lpass.value);
 if(!current)return alert("Invalid login. Please register first.");
 welcome.innerText="Welcome, "+current.name;
 showHistory();
}
```

```
function searchTrains(){
 trainList.innerHTML="";
 let f=origin.value.trim().toLowerCase();
 let d=destination.value.trim().toLowerCase();
 if(!f||!d||!journeyDate.value)return alert("Enter stations and journey date.");
 if(journeyDate.value<new Date().toLocaleDateString("en-CA"))
 return alert("Select a valid journey date.");

 let found=trains.filter(t=>t.from.toLowerCase().includes(f)&&
 t.to.toLowerCase().includes(d));

 if(!found.length){trainList.innerText="No trains found.";return;}

 found.forEach(t=>{
 trainList.innerHTML+=`<div class="train">
 <h3>${t.name}</h3>
 <p>${t.from} → ${t.to}</p>
 <p>Class: ${classType.value} | Fare: ₹${t.fare}</p>
 <p>Available Seats: ${t.seats}</p>
 <button onclick="selectTrain(${t.id})">Select Train</button>
 </div>`;
 });
}

function selectTrain(id){
 selected=trains.find(t=>t.id===id);
 selectedTrain.innerText=selected.name+" | "+selected.from+
 " → "+selected.to+" | "+classType.value;
}

function bookTicket(){
```

```
 if(!current)return alert("Please login first.");
 if(!selected)return alert("Please select a train.");
 if(!journeyDate.value||!passengerName.value.trim()||
 Number(passengerAge.value)<1)return alert("Enter valid passenger details.");
 if(selected.seats<=0)return alert("No seats available.");

 selected.seats--;
 let pnr=String(Math.floor(1000000000+Math.random()*9000000000));
 let status="Confirmed";
 let fare=selected.fare;
 let t={pnr,email:current.email,name:passengerName.value.trim(),
 train:selected.name,from:selected.from,to:selected.to,
 date:journeyDate.value,fare,status,cancelled:false};
 tickets.push(t);
 bookingMsg.innerText="Ticket booked successfully! PNR: "+pnr;
 ticket.innerHTML=`<div class="train" id="printTicket">
 <h3>RailConnect E-Ticket</h3>
 <p>Passenger: ${t.name}</p><p>Train: ${t.train}</p>
 <p>Route: ${t.from} → ${t.to}</p><p>Date: ${t.date}</p>
 <p>Class: ${classType.value}</p><p>PNR: ${pnr}</p>
 <p>Status: ${status} | Fare: ₹${fare}</p>
 <button onclick="window.print()">Print Ticket</button></div>`;
 showHistory();
}

function checkPNR(){
 let p=pnrInput.value.trim();
 let t=tickets.find(x=>x.pnr===p);
 pnrResult.innerText=t?"PNR: "+t.pnr+" | "+t.train+
 " | "+t.date+" | Status: "+(t.cancelled?"Cancelled":t.status):
 "PNR not found.";
}
```

```
function showHistory(){
 let own=tickets.filter(t=>t.email===current.email);
 historyList.innerHTML=own.length?"":"No reservations yet.";
 own.forEach(t=>{
 historyList.innerHTML+=`<div class="train">
 PNR: ${t.pnr} | ${t.train} | ${t.date} | ₹${t.fare}
 | ${t.cancelled?"Cancelled":t.status}</div>`;
 });
}

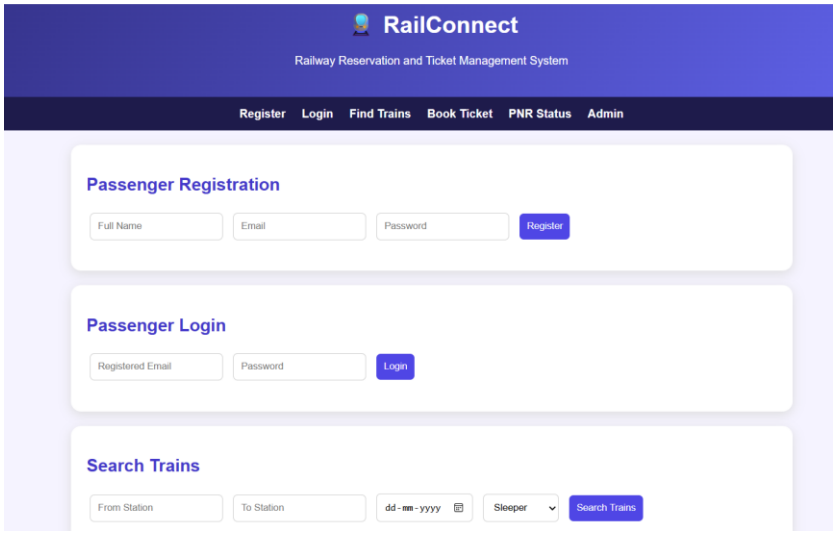
function cancelTicket(){
 let p=cancelPNR.value.trim();
 let t=tickets.find(x=>x.pnr===p);
 if(!t)return alert("Invalid PNR.");
 if(t.cancelled)return alert("Ticket already cancelled.");
 let refund=t.fare*0.8;
 t.cancelled=true;
 let tr=trains.find(x=>x.name===t.train);
 if(tr)tr.seats++;
 refundResult.innerText="Ticket cancelled. Demo refund amount: ₹"+refund+
 " (80% of fare). Actual refund is not processed.";
 if(current)showHistory();
 generateReport();
}

function adminLogin(){
 if(adminName.value==="admin"&&adminPass.value==="admin123"){
 adminPanel.style.display="block";generateReport();
 }else alert("Invalid admin credentials.");
}
```

```
function addTrain(){
 let name=newTrain.value.trim(),from=newFrom.value.trim(),
 to=newTo.value.trim(),fare=Number(newFare.value),seats=Number(newSeats.value);
 if(!name||!from||!to||fare<=0||seats<0)
 return alert("Enter valid train details.");
 trains.push({id:Date.now(),name,from,to,fare,seats});
 alert("Train added successfully.");
 generateReport();
}

function generateReport(){
 report.innerHTML="Total Trains: "+trains.length+
 "<br>Total Bookings: "+tickets.length+
 "<br>Confirmed Tickets: "+tickets.filter(t=>!t.cancelled).length+
 "<br>Cancelled Tickets: "+tickets.filter(t=>t.cancelled).length;
}
</script>
</body>
</html>
```

Output:



Q5): Online flight reservation system

Code:

```
<!DOCTYPE html>
```

```
<html>
<head>
<title>SkyWings Flight Reservation</title>
<style>
*{box-sizing:border-box}
body{font-family:Arial;margin:0;background:#edf5fb;color:#203047}
header{background:linear-gradient(120deg,#082f49,#0284c7);
color:white;text-align:center;padding:28px}
nav{background:#082f49;padding:15px;text-align:center}
nav a{color:white;text-decoration:none;margin:8px;font-weight:bold}
section{background:white;margin:20px auto;padding:22px;
max-width:1000px;border-radius:12px;box-shadow:0 4px 14px #ccdbe7}
h2{color:#0369a1}
input,select,button,textarea{padding:10px;margin:5px;
border:1px solid #cbd5e1;border-radius:6px}
button{background:#0284c7;color:white;border:0;cursor:pointer}
button:hover{background:#075985}
.flight{background:#e0f2fe;padding:16px;margin:12px 0;border-radius:10px}
footer{background:#082f49;color:white;text-align:center;padding:18px}
@media(max-width:600px){
 nav a{display:inline-block;margin:7px}
 input,select,textarea{max-width:100%}
 section{margin:10px;padding:14px}
}
</style>
</head>
<body>

<header>
<h1>✈ SkyWings Airlines</h1>
<p>Your Journey, Our Priority</p>
</header>
```

```
<nav>
<a href="#register">Register</a>
<a href="#login">Login</a>
<a href="#search">Search Flights</a>
<a href="#mytickets">My Tickets</a>
<a href="#status">Booking Status</a>
<a href="#admin">Admin</a>
<a href="#feedback">Feedback</a>
</nav>

<section id="register">
<h2>Customer Registration</h2>
<input id="rname" placeholder="Full Name">
<input id="remail" type="email" placeholder="Email">
<input id="rpass" type="password" placeholder="Password">
<button onclick="register()">Register</button>
<p id="rmsg"></p>
</section>

<section id="login">
<h2>Customer Login</h2>
<input id="lemail" type="email" placeholder="Registered Email">
<input id="lpass" type="password" placeholder="Password">
<button onclick="login()">Login</button>
<button onclick="forgot()">Forgot Password?</button>
<p id="welcome"></p>
</section>

<section id="search">
<h2>Search Flights</h2>
<input id="origin" placeholder="From City">
```

```
<input id="destination" placeholder="Destination">
<input id="date" type="date">
<select id="classType">
<option>Economy</option>
<option>Business</option>
<option>First Class</option>
</select>
<button onclick="searchFlights()">Search Flights</button>
<div id="flightList"></div>
</section>

<section id="booking">
<h2>Book Your Ticket</h2>
<p id="selectedFlight">Select a flight from the search results.</p>
<input id="passenger" placeholder="Passenger Name">
<input id="age" type="number" placeholder="Passenger Age">
<select id="seat">
<option>Window</option>
<option>Aisle</option>
<option>Middle</option>
</select>
<button onclick="bookFlight()">Book Ticket</button>
<p id="bookingMsg"></p>
</section>

<section id="payment">
<h2>Ticket Payment</h2>
<p>Demo payment only. No real money is charged.</p>
<input id="payPNR" placeholder="Enter Booking PNR">
<select id="payMode">
<option>UPI Demo</option>
<option>Card Demo</option>
```

```
<option>Net Banking Demo</option>
</select>
<button onclick="payTicket()">Confirm Demo Payment</button>
<p id="payResult"></p>
</section>

<section id="mytickets">
<h2>My Tickets</h2>
<div id="ticketList">Please login to view tickets.</div>
</section>

<section id="status">
<h2>View Booking Status</h2>
<input id="statusPNR" placeholder="Enter Booking PNR">
<button onclick="checkStatus()">Check Status</button>
<p id="statusResult"></p>
</section>

<section id="cancel">
<h2>Cancel Ticket</h2>
<input id="cancelPNR" placeholder="Enter Booking PNR">
<button onclick="cancelTicket()">Cancel Booking</button>
<p id="cancelResult"></p>
</section>

<section id="print">
<h2>Print E-Ticket</h2>
<input id="printPNR" placeholder="Enter Booking PNR">
<button onclick="printTicket()">View / Print Ticket</button>
<div id="printArea"></div>
</section>
```

```
<section id="admin">
<h2>Administrator Login</h2>
<p>Demo credentials: admin / admin123</p>
<input id="adminName" placeholder="Admin Username">
<input id="adminPass" type="password" placeholder="Admin Password">
<button onclick="adminLogin()">Login as Admin</button>

<div id="adminPanel" style="display:none">
<h3>Add Flight</h3>
<input id="newName" placeholder="Airline / Flight Name">
<input id="newFrom" placeholder="From City">
<input id="newTo" placeholder="To City">
<input id="newFare" type="number" placeholder="Fare">
<input id="newSeats" type="number" placeholder="Available Seats">
<button onclick="addFlight()">Add Flight</button>
<h3>Admin Report</h3>
<div id="report"></div>
<button onclick="generateReport()">Refresh Report</button>
</div>
</section>

<section id="feedback">
<h2>Customer Feedback</h2>
<input id="fbName" placeholder="Your Name">
<select id="rating">
<option>5 Stars</option>
<option>4 Stars</option>
<option>3 Stars</option>
<option>2 Stars</option>
<option>1 Star</option>
</select><br>
<textarea id="fbMessage" placeholder="Write your feedback"></textarea><br>
```

```
<button onclick="feedback()">Submit Feedback</button>
<div id="feedbackList"></div>
</section>

<footer>
<p>SkyWings Airlines | Internet Programming Lab</p>
<p>Contact: support@skywings.example</p>
</footer>

<script>
let users=[],current=null,selected=null,tickets=[],feedbacks=[];
let flights=[
{id:1,name:"Sky Express",from:"Hyderabad",to:"Delhi",fare:4500,seats:5},
{id:2,name:"Air Connect",from:"Chennai",to:"Mumbai",fare:3800,seats:6},
{id:3,name:"Blue Wings",from:"Delhi",to:"Hyderabad",fare:4200,seats:4}
];

function register(){
 let name=rname.value.trim(),email=remail.value.trim().toLowerCase(),pass=rpass.value;
 if(!name||!email.includes("@")||pass.length<4)return alert("Enter valid details.");
 if(users.some(u=>u.email===email))return alert("Email already registered.");
 users.push({name,email,pass});
 rmsg.innerText="Registration successful! Please login.";
}
function login(){
 let email=lemail.value.trim().toLowerCase();
 current=users.find(u=>u.email===email&&u.pass===lpass.value);
 if(!current)return alert("Invalid login. Register first.");
 welcome.innerText="Welcome, "+current.name;
 showTickets();
}
function forgot(){
```

```
 let email=prompt("Enter your registered email:");
 let u=users.find(x=>x.email===email?.trim().toLowerCase());
 alert(u?"Demo: Contact support to reset your password.":"Email not found.");
}
function searchFlights(){
 flightList.innerHTML="";
 let f=origin.value.trim().toLowerCase(),d=destination.value.trim().toLowerCase();
 if(!f||!d||!date.value)return alert("Enter route and travel date.");
 if(date.value<new Date().toLocaleDateString("en-CA"))return alert("Choose a valid date.");
 let list=flights.filter(x=>x.from.toLowerCase().includes(f)&&x.to.toLowerCase().includes(d));
 if(!list.length){flightList.innerText="No flights found.";return;}
 list.forEach(x=>{
 flightList.innerHTML+=`<div class="flight"><h3>${x.name}</h3>
 <p>${x.from} → ${x.to}</p><p>Fare: ₹${x.fare} | Class: ${classType.value}</p>
 <p>Available seats: ${x.seats}</p>
 <button onclick="selectFlight(${x.id})">Select Flight</button></div>`;
 });
}
function selectFlight(id){
 selected=flights.find(x=>x.id===id);
 selectedFlight.innerText=selected.name+" | "+selected.from+" → "+selected.to+
 " | ₹"+selected.fare+" | "+classType.value;
}
function bookFlight(){
 if(!current)return alert("Please login first.");
 if(!selected)return alert("Search and select a flight first.");
 if(!date.value||!passenger.value.trim()||Number(age.value)<1)return alert("Enter passenger details.");
 if(selected.seats<=0)return alert("No seats available.");
 selected.seats--;
 let pnr=String(Math.floor(100000000+Math.random()*900000000));
 tickets.push({pnr,email:current.email,name:passenger.value.trim(),
 flight:selected.name,from:selected.from,to:selected.to,date:date.value,
```

```
 fare:selected.fare,class:classType.value,seat:seat.value,
 status:"Payment Pending",cancelled:false});
 bookingMsg.innerText="Booking created. PNR: "+pnr+". Complete demo payment.";
 showTickets();
}
function payTicket(){
 let t=tickets.find(x=>x.pnr===payPNR.value.trim());
 if(!t)return alert("Invalid PNR.");
 if(t.cancelled)return alert("Ticket is cancelled.");
 if(t.status==="Confirmed")return alert("Already paid.");
 t.status="Confirmed";
 payResult.innerText="Demo payment successful! PNR: "+t.pnr;
 showTickets();generateReport();
}
function showTickets(){
 let list=tickets.filter(t=>t.email===current.email);
 ticketList.innerHTML=list.length?"":"No tickets found.";
 list.forEach(t=>{
 ticketList.innerHTML+=`<div class="flight">${t.flight} | ${t.from} → ${t.to}
 <br>Date: ${t.date} | PNR: ${t.pnr} | ₹${t.fare}
 <br>Status: ${t.status} ${t.cancelled?"(Cancelled)":""}</div>`;
 });
}
function checkStatus(){
 let t=tickets.find(x=>x.pnr===statusPNR.value.trim());
 statusResult.innerText=t?"PNR: "+t.pnr+" | "+t.flight+" | "+t.status:"PNR not found.";
}
function cancelTicket(){
 let t=tickets.find(x=>x.pnr===cancelPNR.value.trim());
 if(!t)return alert("PNR not found.");
 if(t.cancelled)return alert("Already cancelled.");
 t.cancelled=true;t.status="Cancelled";
```

```
 let f=flights.find(x=>x.name===t.flight);if(f)f.seats++;
 cancelResult.innerText="Ticket cancelled. Contact airline for refund processing.";
 if(current)showTickets();generateReport();
}
function printTicket(){
 let t=tickets.find(x=>x.pnr===printPNR.value.trim());
 if(!t||t.cancelled||t.status!=="Confirmed")return alert("Valid confirmed ticket not found.");
 printArea.innerHTML=`<div class="flight"><h3>SkyWings E-Ticket</h3>
 Passenger: ${t.name}<br>Flight: ${t.flight}<br>Route: ${t.from} → ${t.to}
 <br>Date: ${t.date}<br>Class: ${t.class}<br>Seat Preference: ${t.seat}
 <br>PNR: ${t.pnr}<br>Fare: ₹${t.fare}<br>Status: ${t.status}
 <br><button onclick="window.print()">Print</button></div>`;
}
function adminLogin(){
 if(adminName.value==="admin"&&adminPass.value==="admin123"){
 adminPanel.style.display="block";generateReport();
 }else alert("Invalid admin credentials.");
}
function addFlight(){
 let name=newName.value.trim(),from=newFrom.value.trim(),to=newTo.value.trim();
 let fare=Number(newFare.value),seats=Number(newSeats.value);
 if(!name||!from||!to||fare<=0||seats<0)return alert("Enter valid flight details.");
 flights.push({id:Date.now(),name,from,to,fare,seats});
 alert("Flight added successfully.");generateReport();
}
function generateReport(){
 report.innerHTML="Total Flights: "+flights.length+
 "<br>Total Bookings: "+tickets.length+
 "<br>Confirmed: "+tickets.filter(t=>t.status==="Confirmed"&&!t.cancelled).length+
 "<br>Cancelled: "+tickets.filter(t=>t.cancelled).length;
}
function feedback(){
```

```
 if(!fbName.value.trim()||!fbMessage.value.trim())return alert("Enter feedback details.");
 feedbacks.push({name:fbName.value.trim(),rating:rating.value,message:fbMessage.value.trim()});
 feedbackList.innerHTML=feedbacks.map(f=>`<div class="flight"><b>${f.name}</b> -
${f.rating}<p>${f.message}</p></div>`).join("");
}
</script>
</body>
</html>
```

Output:



Q6): Online cake ordering system

Code:

```
<!DOCTYPE html>
<html>
<head>
<title>Cake Ordering System</title>
<style>
body {
  font-family: Arial;
  margin: 0;
  background: #fff3f5;
  color: #4a2525;
}
header {
```

```
  background: #b8325b;
  color: white;
  padding: 18px;
  text-align: center;
}
nav {
  background: #ffe0e8;
  padding: 12px;
  text-align: center;
}
nav a {
  margin: 12px;
  color: #8b2445;
  text-decoration: none;
  font-weight: bold;
}
.hero {
  text-align: center;
  padding: 25px;
  background: #fffaf0;
}
input, select {
  padding: 10px;
  margin: 6px;
  border: 1px solid #dba0b0;
  border-radius: 5px;
}
.cakes {
  display: flex;
  justify-content: center;
  flex-wrap: wrap;
  gap: 20px;
```

```
  padding: 20px;
}
.card {
  background: white;
  width: 200px;
  padding: 15px;
  text-align: center;
  border-radius: 12px;
  box-shadow: 0 3px 10px #d8aab5;
}
.card .image {
  font-size: 65px;
}
button {
  background: #b8325b;
  color: white;
  padding: 10px;
  border: none;
  border-radius: 5px;
  cursor: pointer;
}
button:hover { background: #862442; }
footer {
  background: #4a2525;
  color: white;
  text-align: center;
  padding: 15px;
}
</style>
</head>
<body>
```

```
<header>
  <h1>🍰 Sweet Cake Shop</h1>
  <p>Fresh Cakes for Every Occasion</p>
</header>

<nav>
  <a href="#home">Home</a>
  <a href="#cakes">Cakes</a>
  <a href="#order">Order</a>
  <a href="#login">Login</a>
</nav>

<section class="hero" id="home">
  <h2>Welcome to Sweet Cake Shop</h2>
  <p>Order delicious cakes for birthdays and celebrations.</p>
  <input id="search" placeholder="Search cakes..." oninput="searchCakes()">
</section>

<section class="cakes" id="cakes">
  <div class="card">
    <div class="image">🎂</div>
    <h3>Chocolate Cake</h3>
    <p>Rich chocolate flavor</p>
    <p>₹500</p>
    <button onclick="add('Chocolate Cake',500)">Add to Cart</button>
  </div>

  <div class="card">
    <div class="image">🍓</div>
    <h3>Strawberry Cake</h3>
    <p>Fresh strawberry cream</p>
    <p>₹600</p>
```

```
    <button onclick="add('Strawberry Cake',600)">Add to Cart</button>
  </div>

  <div class="card">
    <div class="image">🍰</div>
    <h3>Vanilla Cake</h3>
    <p>Classic vanilla flavor</p>
    <p>₹400</p>
    <button onclick="add('Vanilla Cake',400)">Add to Cart</button>
  </div>
</section>

<section class="hero" id="order">
  <h2>Your Cart</h2>
  <p id="cart">No cakes selected</p>
  <h3 id="total">Total: ₹0</h3>
  <button onclick="placeOrder()">Place Order</button>
  <p id="result"></p>
</section>

<section class="hero" id="login">
  <h2>Customer Login</h2>
  <input type="email" placeholder="Email" required>
  <input type="password" placeholder="Password" required>
  <button onclick="alert('Demo login only')">Login</button>
</section>

<footer>
  <p>© 2026 Sweet Cake Shop | Online Cake Ordering</p>
</footer>

<script>
```

```
let total = 0;
let items = [];

function add(name, price) {
  items.push(name);
  total += price;
  document.getElementById("cart").innerText = items.join(", ");
  document.getElementById("total").innerText = "Total: ₹" + total;
}

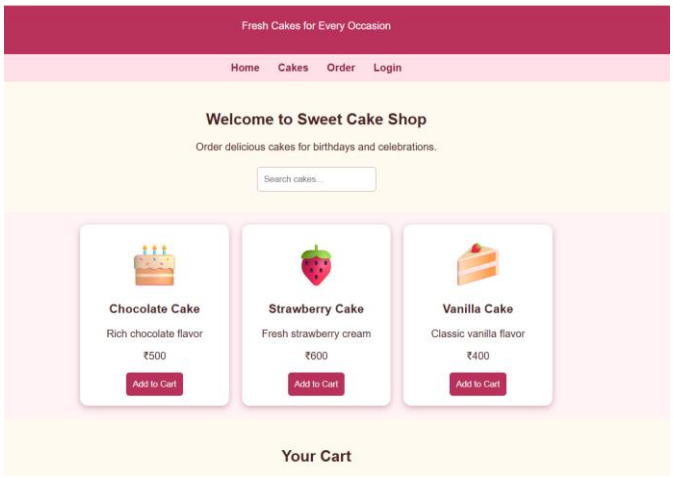
function placeOrder() {
  if (items.length === 0) {
    alert("Please select a cake first!");
  } else {
    document.getElementById("result").innerText =
      "Order placed successfully! Total: ₹" + total;
    items = [];
    total = 0;
    document.getElementById("cart").innerText = "No cakes selected";
    document.getElementById("total").innerText = "Total: ₹0";
  }
}

function searchCakes() {
  let value = document.getElementById("search").value.toLowerCase();
  document.querySelectorAll(".card").forEach(card => {
    card.style.display =
      card.innerText.toLowerCase().includes(value) ? "" : "none";
  });
}
</script>
```

</body>

</html>

Output:



Q7): Online beauty parlour appointment booking system

Code:

<!DOCTYPE html>

<html lang="en">

<head>

<meta charset="UTF-8">

<meta name="viewport" content="width=device-width, initial-scale=1.0">

<title>Glow Beauty Parlour</title>

<style>

*{box-sizing:border-box}

body{margin:0;font-family:Arial;background:#fff7fb;color:#352536}

header{background:#702963;color:white;padding:20px 8%;

display:flex;justify-content:space-between;align-items:center;flex-wrap:wrap}

header h2{margin:0}

nav a{color:white;text-decoration:none;margin:10px;font-weight:bold}

.hero{text-align:center;padding:45px 15px;background:#f2d7ed}

.hero h1{color:#702963;font-size:34px}

button{background:#702963;color:white;border:0;padding:11px 17px;

border-radius:6px;cursor:pointer}

section{padding:25px 8%}

```
.services{display:flex;gap:18px;flex-wrap:wrap}
.card{background:white;padding:20px;border-radius:12px;
box-shadow:0 3px 12px #e7cddd;flex:1;min-width:190px;text-align:center}
.card h3{color:#702963}
.booking{background:white;padding:22px;border-radius:12px;
box-shadow:0 3px 12px #e7cddd;max-width:600px;margin:auto}
input,select,textarea{width:100%;padding:11px;margin:8px 0 14px;
border:1px solid #d9b9d1;border-radius:6px}
#message{color:#702963;font-weight:bold}
table{width:100%;border-collapse:collapse;background:white}
th,td{padding:12px;border:1px solid #ead7e5;text-align:left}
th{background:#702963;color:white}
footer{text-align:center;background:#702963;color:white;padding:18px}
</style>
</head>
<body>

<header>
  <h2>✨ Glow Beauty Parlour</h2>
  <nav>
    <a href="#home">Home</a>
    <a href="#services">Services</a>
    <a href="#booking">Book Appointment</a>
    <a href="#reservations">Reservations</a>
  </nav>
</header>

<div class="hero" id="home">
  <h1>Feel Beautiful, Look Beautiful</h1>
  <p>Professional beauty services, just one appointment away.</p>
  <a href="#booking"><button>Book Now</button></a>
</div>
```

```
<section id="services">
  <h2>Our Beauty Services</h2>
  <div class="services">
    <div class="card">
      <h3>Hair Styling</h3>
      <p>Haircut, styling and hair care</p>
      <h3>₹500</h3>
      <button onclick="chooseService('Hair Styling')">Select</button>
    </div>
    <div class="card">
      <h3>Facial</h3>
      <p>Skin care and facial treatments</p>
      <h3>₹800</h3>
      <button onclick="chooseService('Facial')">Select</button>
    </div>
    <div class="card">
      <h3>Makeup</h3>
      <p>Party and special occasion makeup</p>
      <h3>₹1500</h3>
      <button onclick="chooseService('Makeup')">Select</button>
    </div>
  </div>
</section>

<section id="booking">
  <h2 style="text-align:center">Book Your Appointment</h2>
  <div class="booking">
    <form id="bookingForm">
      <label>Customer Name</label>
      <input id="name" required placeholder="Enter your name">
```

```
<label>Email Address</label>
<input id="email" type="email" required placeholder="Enter email">

<label>Beauty Service</label>
<select id="service" required>
  <option value="">Select Service</option>
  <option>Hair Styling</option>
  <option>Facial</option>
  <option>Makeup</option>
</select>

<label>Select Stylist</label>
<select id="stylist" required>
  <option value="">Choose Stylist</option>
  <option>Ananya</option>
  <option>Priya</option>
  <option>Meera</option>
</select>

<label>Appointment Date</label>
<input id="date" type="date" required>

<label>Appointment Time</label>
<select id="time" required>
  <option value="">Choose Time</option>
  <option>10:00 AM</option>
  <option>11:00 AM</option>
  <option>12:00 PM</option>
  <option>2:00 PM</option>
  <option>3:00 PM</option>
  <option>4:00 PM</option>
</select>
```

```
      <label>Payment Method</label>
      <select id="payment">
        <option>Pay at Salon</option>
        <option>PayPal (Demo)</option>
        <option>Stripe (Demo)</option>
      </select>

      <button type="submit">Confirm Appointment</button>
      <p id="message"></p>
    </form>
  </div>
</section>

<section id="reservations">
  <h2>My Reservations</h2>
  <table>
    <thead>
      <tr><th>Customer</th><th>Service</th><th>Date</th>
      <th>Time</th><th>Action</th></tr>
    </thead>
    <tbody id="reservationList"></tbody>
  </table>
</section>

<footer>
  <p>© 2026 Glow Beauty Parlour | Beauty • Style • Confidence</p>
</footer>

<script>
const dateInput = document.getElementById("date");
dateInput.min = new Date().toISOString().split("T")[0];
```

```
function chooseService(service) {
  document.getElementById("service").value = service;
  document.getElementById("booking").scrollIntoView();
}

document.getElementById("bookingForm").addEventListener("submit", function(e) {
  e.preventDefault();

  const name = document.getElementById("name").value;
  const service = document.getElementById("service").value;
  const date = dateInput.value;
  const time = document.getElementById("time").value;

  const row = document.createElement("tr");
  [name, service, date, time].forEach(value => {
    const cell = document.createElement("td");
    cell.textContent = value;
    row.appendChild(cell);
  });

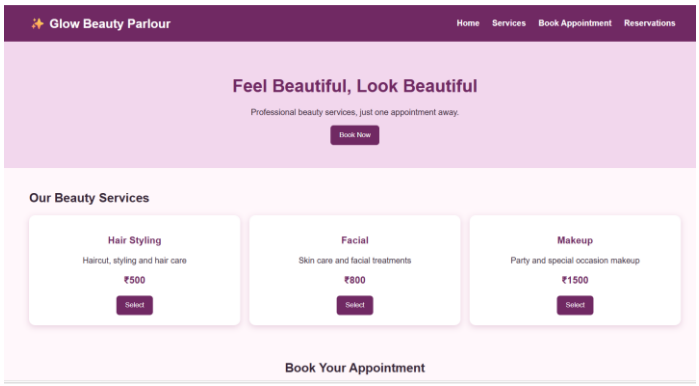
  const action = document.createElement("td");
  const cancel = document.createElement("button");
  cancel.textContent = "Cancel";
  cancel.onclick = () => row.remove();
  action.appendChild(cancel);
  row.appendChild(action);

  document.getElementById("reservationList").appendChild(row);
  document.getElementById("message").textContent =
    "Appointment booked successfully!";
  this.reset();
```

```
});
</script>

</body>
</html>
```

Output:



Q8): MedCare Hospital Appointment Booking System

Code:

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>MedCare Hospital</title>

<style>
*{box-sizing:border-box}
body{margin:0;font-family:Arial,sans-serif;background:#f2f8fc;color:#243746}
header{background:#087cae;color:white;padding:18px 7%;
display:flex;justify-content:space-between;align-items:center;flex-wrap:wrap}
header h2{margin:0}
nav a{color:white;text-decoration:none;margin:10px;font-weight:bold}
```

```
.hero{text-align:center;padding:45px 15px;background:#dff3ff}
.hero h1{color:#087cae;font-size:34px}
button{background:#087cae;color:white;border:0;padding:11px 16px;
border-radius:5px;cursor:pointer}
button:hover{background:#055b82}
section{padding:25px 7%}
.search{max-width:600px;margin:auto}
input,select{width:100%;padding:11px;margin:8px 0 14px;
border:1px solid #c5d9e5;border-radius:5px}
.doctors{display:flex;gap:18px;flex-wrap:wrap}
.card{background:white;padding:20px;border-radius:10px;
box-shadow:0 3px 12px #d7e6ef;flex:1;min-width:190px;text-align:center}
.card h3{color:#087cae}
.booking{max-width:600px;margin:auto;background:white;padding:22px;
border-radius:10px;box-shadow:0 3px 12px #d7e6ef}
table{width:100%;border-collapse:collapse;background:white}
th,td{padding:11px;border:1px solid #d4e3ec;text-align:left}
th{background:#087cae;color:white}
#message{color:#087cae;font-weight:bold}
footer{text-align:center;background:#075477;color:white;padding:18px}
</style>
</head>

<body>

<header>
  <h2>✚ MedCare Hospital</h2>
  <nav>
    <a href="#home">Home</a>
    <a href="#doctors">Doctors</a>
    <a href="#booking">Book Appointment</a>
    <a href="#history">My Appointments</a>
```

```
  </nav>
</header>

<div class="hero" id="home">
  <h1>Your Health, Our Priority</h1>
  <p>Find experienced doctors and book your appointment online.</p>
  <a href="#booking"><button>Book Appointment</button></a>
</div>

<section id="doctors">
  <h2>Our Specialist Doctors</h2>
  <div class="doctors">
    <div class="card">
      <h3>Dr. Rahul</h3>
      <p>Cardiologist</p>
      <p>Consultation: ₹500</p>
      <button onclick="selectDoctor('Dr. Rahul')">Book Now</button>
    </div>
    <div class="card">
      <h3>Dr. Priya</h3>
      <p>Dermatologist</p>
      <p>Consultation: ₹400</p>
      <button onclick="selectDoctor('Dr. Priya')">Book Now</button>
    </div>
    <div class="card">
      <h3>Dr. Arjun</h3>
      <p>General Physician</p>
      <p>Consultation: ₹300</p>
      <button onclick="selectDoctor('Dr. Arjun')">Book Now</button>
    </div>
  </div>
</section>
```

```
<section id="booking">
  <h2 style="text-align:center">Book Your Appointment</h2>
  <div class="booking">
    <form id="appointmentForm">
      <label>Patient Name</label>
      <input id="patient" required placeholder="Enter patient name">

      <label>Email Address</label>
      <input id="email" type="email" required placeholder="Enter email">

      <label>Hospital</label>
      <input id="hospital" required placeholder="Enter hospital name">

      <label>Doctor</label>
      <select id="doctor" required>
        <option value="">Select Doctor</option>
        <option>Dr. Rahul</option>
        <option>Dr. Priya</option>
        <option>Dr. Arjun</option>
      </select>

      <label>Specialty</label>
      <select id="specialty" required>
        <option value="">Select Specialty</option>
        <option>Cardiology</option>
        <option>Dermatology</option>
        <option>General Physician</option>
      </select>

      <label>Appointment Date</label>
      <input id="date" type="date" required>
```

```
      <label>Appointment Time</label>
      <select id="time" required>
        <option value="">Select Time</option>
        <option>10:00 AM</option>
        <option>11:00 AM</option>
        <option>12:00 PM</option>
        <option>2:00 PM</option>
        <option>3:00 PM</option>
        <option>4:00 PM</option>
      </select>

      <button type="submit">Confirm Appointment</button>
      <p id="message"></p>
    </form>
  </div>
</section>

<section id="history">
  <h2>Appointment History</h2>
  <table>
    <thead>
      <tr>
        <th>Patient</th><th>Doctor</th><th>Date</th>
        <th>Time</th><th>Action</th>
      </tr>
    </thead>
    <tbody id="appointmentList"></tbody>
  </table>
</section>

<footer>
```

```
  <p>© 2026 MedCare Hospital | Caring for Your Health</p>
</footer>

<script>
const dateInput = document.getElementById("date");
dateInput.min = new Date().toISOString().split("T")[0];

function selectDoctor(name) {
  document.getElementById("doctor").value = name;
  document.getElementById("booking").scrollIntoView();
}

document.getElementById("appointmentForm").addEventListener("submit", function(e) {
  e.preventDefault();

  const patient = document.getElementById("patient").value;
  const doctor = document.getElementById("doctor").value;
  const date = dateInput.value;
  const time = document.getElementById("time").value;

  const row = document.createElement("tr");

  [patient, doctor, date, time].forEach(value => {
    const cell = document.createElement("td");
    cell.textContent = value;
    row.appendChild(cell);
  });

  const action = document.createElement("td");
  const cancel = document.createElement("button");
  cancel.textContent = "Cancel";
  cancel.onclick = () => row.remove();
```

```
  action.appendChild(cancel);
  row.appendChild(action);

  document.getElementById("appointmentList").appendChild(row);
  document.getElementById("message").textContent =
    "Appointment booked successfully!";
  this.reset();
});
</script>

</body>
</html>
```

Output:

